

Poetic Personality Chatbot – Project

Documentation

By NABIKETH THALARI

Introduction:

This project is a creative chatbot that turns any user input-whether it's a question, thought, or statement-into a short, poetic response. It's built using Google's Gemini API (Flash version) and designed to simulate a poetic personality through prompt engineering. The primary goal was to explore AI's creative capabilities while offering users an emotionally rich, lyrical interaction.

Steps to Build the Chatbot:

1. API Setup:

- I installed the google-genai package using pip.
- I retrieved the Gemini API key securely via Google Colab's userdata.get() method.
- I initialized the Gemini client with genai.Client() and created a chat instance using the "gemini-2.0-flash" model.

2. Prompt Engineering:

- I crafted a specific prompt instructing the model to always reply in poetic form.
- I ensured that even serious or technical questions would result in a poem.

3. Output Display:

- I used a custom stream() function to print responses word-by-word with a short delay, giving the feel of a chatbot "thinking".

Challenges Faced:

- **API key handling:** I did silly mistakes while referencing API key.
- **Response Formatting:** At times, the response wasn't poetic enough. This was solved by fine-tuning the system prompt to be more assertive about poetic output.
- **Rate Limits:** Occasionally, hitting the API too quickly resulted in errors. I added error-handling to inform users gracefully.

Conclusion:

This project taught me how to interface with LL APIs like Gemini and shaped my understanding of prompt engineering's influence on output style. Beyond technical implementation, it deepened my appreciation for AI's capacity to simulate creativity and emotion through controlled interaction.

Reflection:

The poetic personality of the chatbot was greatly influenced by prompt engineering. At first, the model gave generic responses that were frequently simple or factual. The chatbot started producing brief verses, rhyming stanzas, and metaphorical reflections after the prompt was modified to specifically call for poetic responses, regardless of the input. It was striking how a minor phrasing alteration could have a significant impact on the output's tone, organization, and emotion. Conciseness was ensured by adding phrases like "short and sweet," while "respond with a poem" promoted imaginative formatting.

This experience demonstrated that while large language models like Gemini are lacking human-level creativity, they can effectively simulate it when given clear instructions. Despite being entirely algorithmic, the poetic chatbot frequently surprised me with the depth of emotion or amusing tone it might lend to seemingly uninteresting topics.

In the end, I noticed that prompt engineering is a creative art as well as a technical ability. It establishes the line separating a lifeless bot from one that has a nearly human feel. This project confirmed that we, as designers, are AI's muses and that its "creativity" is an echo of our commands.